07/02/2025

Code-A

Aakash
Medical | IIT-JEE | Foundations

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**MM : 702**

CHEMISTRY 12 MOCK TEST

**Time : 180 Min.**

## SECTION A

**1.** If the mass percentage of urea in an aqueous solution is 20% then molality of the solution will be

(1) 5.25 m

(2) 2.35 m

(3) 4.17 m

(4) 4.31 m

**2.** Concentrated aqueous sulphuric acid is 98% $H_2SO_4$ by mass. How many grams of concentrated sulphuric acid solution should be used to prepare
250 ml of 2.0 M $H_2SO_4$?

(1) 98 g

(2) 50 g

(3) 110 g

(4) 100 g

**3.** The concentration term(s) which depends on temperature is/are

(1) Molarity

(2) Weight volume percent

(3) Normality

(4) All of these

**4.** Van't Hoff factor of 0.1 molal $CaCl_2$ solution will be

(1) 2

(2) 4

(3) 1

(4) 3

**5.** (Ethanol + water) mixture can form

(1) Minimum boiling azeotrope having same composition in liquid and vapour phase

(2) Maximum boiling azeotrope which boils at a constant temperature

(3) Maximum boiling azeotrope which shows a large positive deviation from Raoult's law

(4) Minimum boiling azeotrope which shows a large negative deviation from Raoult's law

**6.** Which is an incorrect expression for mixing of pure components to form an ideal solution?

(1) $\Delta_{mix}H = 0$

(2) $\Delta_{mix}V = 0$

(3) $\Delta_{mix}G = 0$

(4) Both (2) and (3)

**7.** Given below are two statements one is labelled as Assertion (A) and other is labelled as Reason (R).
**Assertion (A):** Azeotropes boil at a constant temperature.
**Reason (R):** The solutions which show a large positive deviation from Raoult's law form minimum boiling azeotrope at a specific composition.
In the light of above two statements mark the correct statement.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

**8.** An aqueous solution of a nonvolatile solute boils at 101.20°C. At what temperature the solution will freeze? (Given: $K_b = 0.512$ K kg mol$^{-1}$ and $K_f = 1.86$ K kg mol$^{-1}$)

(1) –2.17°C

(2) –1.2°C

(3) –3.46°C

(4) –4.36°C

**9.** Select the incorrect statement regarding ideal solution of components A and B

(1) Intermolecular attractive force between A – A and B – B are equal to those between A – B

(2) $\Delta_{mix}H = 0$ at constant T and P

(3) $\Delta_{mix}V = 0$ at constant T and P

(4) $\Delta_{mix}S = 0$ at constant T and P

**10.** 0.5 M solution of $MgCl_2$ will have the same osmotic pressure as (consider complete dissociation of electrolytes)

(1) 1 M NaCl

(2) 1.5 M KCl

(3) 1.5 M urea

(4) 1 M $AlCl_3$

**11.** If 0.1 m aqueous solution of calcium phosphate is 80% dissociated then the freezing point of the solution will be ($K_f$ of water = 1.86 K kg $mol^{-1}$)

(1) –0.78°C

(2) –1.25°C

(3) –1.75°C

(4) –2.5°C

**12.** 100 ml of 2 M $H_2SO_4$ solution is diluted by adding 400 ml water. The molarity of the resultant solution becomes

(1) 0.4 M

(2) 0.6 M

(3) 0.75 M

(4) 0.2 M

**13.** The mass of urea to be dissolved in 100 g of water ($K_b$ = 0.52 K kg $mol^{-1}$) to prepare a solution having boiling point 101°C, is

(1) 10.26 g

(2) 4.2 g

(3) 11.54 g

(4) 14.66 g

**14.** The molecular weight of benzoic acid in benzene is determined by depression in the freezing point method corresponds to

(1) Tetramerisation of benzoic acid

(2) Dimerisation of benzoic acid

(3) Trimerisation of benzoic acid

(4) Ionisation of benzoic acid

**15.** Given below are two statements one is assertion (A) other is reason (R)
**Assertion (A):** Aquatic species are more comfortable in cold water than in warm water.
**Reason (R):** Solubility of oxygen in water decreases with increase in temperature.
In the light of above statements choose the correct answer.

(1) Both assertion and reason are correct statements, and reason is the correct explanation of the assertion.

(2) Both assertion and reason are correct statements, but reason is not the correct explanation of the assertion.

(3) Assertion is correct, but reason is wrong statement.

(4) Assertion is wrong, but reason is correct statement.

**16.** The reduction potential of hydrogen electrode at 298 K and pH = 10 is

(1) 0.591 V

(2) –0.059 V

(3) –0.591 V

(4) Zero

**17.** A : At equilibrium $E_{cell}$ becomes zero.
R : At equilibrium reduction potential of both the electrodes becomes equal.

(1) Both assertion (A) and reason (R) are correct statements, and reason is the correct explanation of the assertion.

(2) Both assertion (A) and reason (R) are correct statements, but reason is not the correct explanation of the assertion.

(3) Assertion (A) is correct, but reason (R) is wrong statement.

(4) Assertion (A) is wrong but reason (R) is correct statement.

**18.** Which of the following is true?

(1) $\Lambda_m^o KCl - \Lambda_m^o NaCl = \Lambda_m^o KBr - \Lambda_m^o NaBr$

(2) $\Lambda_m^o KCl + \Lambda_m^o NaCl = \Lambda_m^o KBr + \Lambda_m^o NaBr$

(3) $\Lambda_m^o KCl - \Lambda_m^o NaBr = \Lambda_m^o KBr - \Lambda_m^o NaCl$

(4) All of these

**19.** The limiting ionic molar conductivities of $K^+$, $Al^{3+}$, $SO_4^{2-}$ ions are x, y and z S $cm^2$ $mol^{-1}$ respectively. The $\lambda_m^o$ value for potash alum is

(1) (2x + y + 4z) S $cm^2$ $mol^{-1}$

(2) (2x + 2y + 4z) S $cm^2$ $mol^{-1}$

(3) (2x + 2y – 4z) S $cm^2$ $mol^{-1}$

(4) $\left(\frac{x}{4} + \frac{3y}{4} + z\right)$ S $cm^2$ $mol^{-1}$

**20.** The weight of copper (At wt = 63.5) displaced by a quantity of electricity which displaces 11200 mL of $O_2$ gas at STP will be

(1) 31.75 g

(2) 63.5 g

(3) 95.25 g

(4) 127 g

**21.** The limiting molar conductivities at infinite dilution for KOH, $KNO_3$ and $NH_4NO_3$ respectively are 239, 125 and 128 S $cm^2 mol^{-1}$. If molar conductivity of 0.2 M of $NH_4OH$ solution is 14 S $cm^2$ $mol^{-1}$ then degree of dissociation of $NH_4OH$ is nearly

(1) 0.01

(2) 0.02

(3) 0.06

(4) 0.15

**22.** The standard reduction potential E° of the following systems are:

| | System | E°(volts) |
|---|---|---|
| (i) | $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ | 1.51 |
| (ii) | $Sn^{4+} + 2e^- \rightarrow Sn^{2+}$ | 0.15 |
| (iii) | $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$ | 1.33 |
| (iv) | $Ce^{4+} + e^- \rightarrow Ce^{3+}$ | 1.61 |

The oxidising power of the various species decreases in the order.

(1) $Ce^{4+} > Cr_2O_7^{2-} > Sn^{4+} > MnO_4^-$

(2) $Ce^{4+} > MnO_4^- > Cr_2O_7^{2-} > Sn^{4+}$

(3) $Cr_2O_7^{2-} > Sn^{4+} > Ce^{4+} > MnO_4^-$

(4) $MnO_4^- > Ce^{4+} > Sn^{4+} > Cr_2O_7^{2-}$

**23.** Which among the following metal will not liberate hydrogen when reacted with dilute HCl solution?

(1) Na

(2) Mg

(3) Cu

(4) Zn

**24.** During recharging of lead storage battery, the reaction taking place at cathode is

(1) $Pb(s) + SO_4^{2-}(aq) \rightarrow PbSO_4(s) + 2e^-$

(2) $PbSO_4(s) + 2H_2O\ (l) \rightarrow PbO_2\ (s) + SO_4^{2-}(aq) + 4H^+(aq) + 2e^-$

(3) $PbSO_4(s) + 2e^- \rightarrow Pb(s) + SO_4^{2-}\ (aq)$

(4) $PbO_2(s) + SO_4^{2-}(aq) + 4H^+(aq) + 2e^- \rightarrow PbSO_4(s) + 2H_2O(l)$

**25.** Number of Faradays required to convert 1 mol of $ClO_3^-$ to $Cl^-$ is

(1) 6F

(2) 3F

(3) 4F

(4) 2F

**26.** Intercept of graph which is plotted between $\Lambda_m$ and $c^{1/2}$ for strong electrolyte is ($\Lambda_m$ plotted on y-axis)

(1) $\Lambda_m^0$

(2) $\sqrt{\kappa}$

(3) $\kappa$

(4) $\frac{1}{\Lambda_m^0}$

**27.** The reduction potential for two half reactions are given :

$2H_2O(l) + 2e^{(-)} \rightarrow H_2(g) + 2OH^{(-)}(aq); E° = -0.8\ V$

$Na^{\oplus}(aq) + e^{(-)} \rightarrow Na(s); E° = -2.7V$

Aqueous sodium ion cannot be reduced to solid sodium in an electrolytic cell because

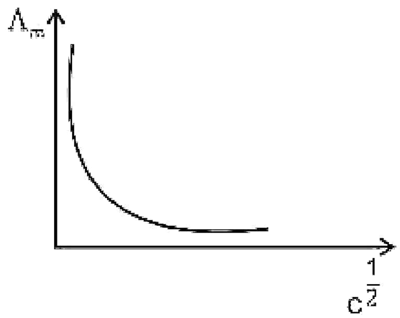
(1) $Na^{(+)}(aq)$ will combine with $OH^{(-)}(aq)$ to form NaOH(s)

(2) $H_2O(l)$ is more easily reduced than aqueous sodium ion

(3) NaOH is a strong base

(4) $H_2$ is gas whereas Na is solid at room temperature

**28.**




Which of the following electrolyte shows the above variation of $\Lambda_m$ vs $C^{\frac{1}{2}}$?

(1) NaCl

(2) $MgCl_2$

(3) $MgSO_4$

(4) $CH_3COOH$

**29.** If the molar conductance of 0.04 M solution of weak monobasic acid is 10 S $cm^2\ mol^{-1}$ and at infinite dilution is 500 S $cm^2\ mol^{-1}$ then the dissociation constant of the acid will be

(1) $8 \times 10^{-4}$

(2) $1.6 \times 10^{-5}$

(3) $2.4 \times 10^{-6}$

(4) $4.0 \times 10^{-4}$

**30.** Mass of copper deposited at the cathode when the aq solution of $CuSO_4$ is electrolysed for 9.65 minute with a current of 10 Ampere is

(1) 30 × 63.5 g

(2) 3 × 63.5 g

(3) 0.03 × 63.5 g

(4) 63.5 g

**31.** The slope of line plotted between ln k versus $\left(\frac{1}{T}\right)$ for Arrhenius equation is given by

(1) $\frac{-E_a}{2.303R}$

(2) $-\frac{E_a}{R}$

(3) $\frac{-2.303E_a}{R}$

(4) $\frac{-R}{E_a}$

**32.** A zero order reaction has initial concentration of reactant 0.1 M and rate constant 0.002 mol $L^{-1}$ $s^{-1}$. Left concentration of reactant after 30 seconds is

(1) 0.02 M

(2) 0.08 M

(3) 0.04 M

(4) 0.03 M

**33.** A hypothetical reaction $A_2 + B_2 \rightarrow 2$ AB follows the mechanism given below:
$A_2 \rightleftharpoons A + A$ (fast)
$A + B_2 \longrightarrow AB + B$ (slow)
$A + B \longrightarrow AB$ (fast)
The order of overall reaction is

(1) 2

(2) 3

(3) 3.5

(4) 1.5

**34.** When initial concentration of a reactant is doubled in a reaction, its Half-life period also gets doubled. The order of the reaction is

(1) First

(2) Zero

(3) Third

(4) Second

**35.** The rate of reaction A $\rightarrow$ B, is $5.0 \times 10^{-2}$ mol $L^{-1}$ $s^{-1}$. The time in which the concentration of reactants changed from 0.04 mol $L^{-1}$ to 0.03 mol $L^{-1}$ is

(1) 0.2 s

(2) 2.0 s

(3) 0.8 s

(4) 1.3 s

**36.** If the rate constant k for the decomposition of HI is related with temperature as $10^{10}.e^{-\frac{18000}{T}}$, then the activation energy is

(1) $-18000 \times R$

(2) $-\frac{18000}{R}$

(3) $10^{10}R$

(4) $18000 \times R$

**37.** The rate constant of the reaction, A $\rightarrow$ 2 B is $6.93 \times 10^{-2}$ $s^{-1}$. If initial concentration of A is 2 M then the time in which B becomes 2 M is

(1) 10 s

(2) 20 s

(3) 10 min

(4) 20 min

**38.** The unit of rate constant for zero order reaction is

(1) mol $time^{-1}$

(2) L $time^{-1}$

(3) mol $L^{-1}$ $time^{-1}$

(4) L $mol^{-1}$ $time^{-1}$

**39.** A first order reaction has a specific reaction rate of $10^{-3}$ $s^{-1}$. How much time will it take for 40 g of the reactant to reduce to 10 g?

(1) $2 \times 10^{-3} \times 0.693$ s

(2) $\frac{2 \times 0.693}{10^{-3}}$ s

(3) $\frac{4 \times 0.693}{10^{-3}}$ s

(4) $\frac{0.693}{4 \times 10^{-3}}$ s



**40.** For a first order reaction, the time required for 99.9% completion is 100 min. What will be the rate constant for the reaction?

(1) $1.25 \times 10^{-4}$ $s^{-1}$

(2) $1.15 \times 10^{-3}$ $s^{-1}$

(3) $2.2 \times 10^{-2}$ $s^{-1}$

(4) $4.5 \times 10^{-2}$ $s^{-1}$

**41.** If in a first order reaction, the concentration of reactant is reduced to 50% in 24 hours then half life period of the reaction is

(1) 6 hours

(2) 8 hours

(3) 24 hours

(4) 48 hours

**42.** Unit of rate constant for a first order reaction is

(1) $s^{-1}$

(2) mol $L^{-1}$ $s^{-1}$

(3) $mol^{-1}$ L $s^{-1}$

(4) $mol^{-1}$ $L^2$ $s^{-1}$

**43.** Which of the following statements for the order of a reaction is incorrect?

(1) Order of reaction is always a whole number

(2) Order can be determined only experimentally

(3) Order is not influenced by stoichiometric coefficient of the reactants

(4) Order of reaction is sum of powers of the concentration terms present in the expression of rate law

**44.** If $t_{1/2} \propto [R]_0$ then order of reaction is

(1) First

(2) Second

(3) Third

(4) Zero

**45.** The rate of a first order reaction is $2.5 \times 10^{-3}$ mol $L^{-1}$ $min^{-1}$ at 0.25 M concentration of reactant. The half life (in min) of the reaction is

(1) 69.3

(2) 43.5

(3) 76.6

(4) 38.7

**46.** $NCl_5$ does not exist. It is due to

(1) Small size

(2) High covalency

(3) Non-availability of d-orbitals

(4) $d\pi - d\pi$ bond formation

**47.** During the oxidation of iodine by concentrated nitric acid, the oxidation state of iodine atom changes from

(1) Zero to –1

(2) Zero to +5

(3) Zero to +1

(4) Zero to +3

**48.** Copper on reaction with conc. $H_2SO_4$ gives

(1) $CuSO_4$, $SO_2$ and $H_2O$

(2) $CuSO_4$ and $H_2$

(3) $CuSO_4$ and $SO_3$

(4) CuO, $SO_2$ and $H_2O$

**49.** Match processes in List-I with their Product in List-II.

| | **List-I** | | **List-II** |
|---|---|---|---|
| (a) | Heating barium azide | (i) | Chlorine |
| (b) | Haber's process | (ii) | Sulphuric acid |
| (c) | Contact process | (iii) | Ammonia |
| (d) | Deacon's process | (iv) | Pure Nitrogen |

Which of the following is the correct option?

(1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(2) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(3) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

**50.** The correct order of bond dissociation enthalpy of halogens is

(1) $F_2 > Cl_2 > Br_2 > I_2$

(2) $Cl_2 > Br_2 > F_2 > I_2$

(3) $F_2 > Br_2 > I_2 > Cl_2$

(4) $F_2 > Cl_2 > I_2 > Br_2$

**51.** Laboratory grade nitric acid contains nearly

(1) 68% $HNO_3$ by mass

(2) 96% $HNO_3$ by mass

(3) 98% $HNO_3$ by mass

(4) 32% $HNO_3$ by mass



**52.** Number of P-OH bond(s) present in pyrophosphoric acid is

(1) 1

(2) 2

(3) 3

(4) 4

**53.** Which gas is evolved when Zinc reacts with dilute nitric acid?

(1) $NO_2$

(2) NO

(3) $N_2O$

(4) $N_2$

**54.** Incorrect match of name of oxyacids of phosphorous is

(1) $H_3PO_2$ : Phosphinic acid

(2) $H_3PO_3$ : Phosphonic acid

(3) $H_4P_2O_6$ : Hypophosphorous acid

(4) $H_4P_2O_5$ : Pyrophosphorous acid

**55.** The caro's acid is

(1) $H_2SO_3$

(2) $H_2SO_5$

(3) $H_2S_2O_7$

(4) $H_2S_2O_8$

**56.** Hydrolysis of which of the following xenon fluoride is not a redox reaction?

(1) $XeF_2$

(2) $XeF_4$

(3) $XeF_6$

(4) All of these

**57.** Which among the following is a paramagnetic compound?

(1) $N_2O$

(2) NO

(3) $N_2O_3$

(4) $N_2O_4$

**58.** Which of the following does not contain S-S linkage?

(1) $H_2S_2O_6$

(2) $H_2S_2O_3$

(3) $H_2S_2O_8$

(4) $H_2S_4O_6$

**59.** Among the following, the correct order of acidity is

(1) $HClO_4 > HClO_3 > HClO_2$

(2) $HClO_3 > HClO_4 > HClO_2$

(3) $HClO_2 > HClO_3 > HClO_4$

(4) $HClO_4 > HClO_2 > HClO_3$

**60.** Oxoacids of sulphur which has peroxy linkage, is

(1) $H_2S_2O_7$

(2) $H_2S_2O_8$

(3) $H_2S_2O_6$

(4) $H_2S_2O_5$

**61.** Which of the following has strongest 'C – O' bond?

(1) $[Mn(CO)_6]^+$

(2) $[Ni(CO)_4]$

(3) $[Co(CO)_4]^-$

(4) $[Fe(CO)_4]^{2-}$

**62.** If a complex $[M(H_2O)_6]^{2+}$ absorbs energy corresponding to λ nm, then the wavelength (in nm) absorbed by tetrahedral, complex $[M(H_2O)_4]^{2+}$ will be

(1) $\frac{9}{4}\lambda$

(2) $\frac{4}{9}\lambda$

(3) $\frac{5}{9}\lambda$

(4) $\frac{1}{\lambda}$

**63.** Which of the following complex ions is not an inner orbital complex?

(1) $[Co(NH_3)_6]^{3+}$

(2) $[CoF_6]^{3-}$

(3) $[Fe(CN)_8]^{3-}$

(4) $[Fe(CN)_8]^4$

**64.** Which of the following square planar complexes does not show geometrical isomerism?

(1) $[Pt(NH_3)_3Cl]$

(2) $[Pt(gly)_2]$

(3) $[Pt(NH_3)_2BrCl]$

(4) $[Pt(NH_3)(NO_2)BrCl]$

**65.** The total number of ions produced from the complex $K_4[Fe(CN)_6]$ in solution is

(1) 3

(2) 4

(3) 5

(4) 6

**66.** Number of stereoisomers in $Ma_3b_3$ type of coordination compound is

(1) 2

(2) 3

(3) 4

(4) 8

**67.** Which compound is paramagnetic with respect to four electrons?

(1) $[Mn(CN_6]^{3-}$

(2) $[Fe(CN)_6]^{3-}$

(3) $[FeF_6]^{3-}$

(4) $[MnCl_6]^{3-}$

**68.** For $[Fe(CN)_6]^{-4}$, $[Fe(H_2O)_6]^{3+}$ and $[Fe\ F_6]^{-4}$ select the correct statements.
a. All have identical geometry.
b. All are paramagnetic.
c. $[Fe(CN)_6]^{-4}$ is diamagnetic but $[Fe(H_2O)_6]^{+3}$ and $[Fe\ F_6]^{-4}$ are paramagnetic.
d. All are diamagnetic.

(1) Only a

(2) Only b

(3) Only a & c

(4) Only a & d

**69.** Optical isomerism is shown by which of the following complex?

(1) $[PtCl_4]^{2-}$

(2) trans-$[CrCl_2(ox)_2]^{3-}$

(3) $[Co(en)_3]^{3+}$

(4) trans-$[Fe(NH_3)_2(CN)_4]^{-}$

**70.** Among the following complex entities, which will exhibit maximum ion conductivity?

(1) $[Fe(CO)_5]$

(2) $[Ni(CO)_4]$

(3) $[Co(NH_3)_6]Cl_3$

(4) $K_4[Fe(CN)_6]$

**71.** Which one of the following is an outer orbital complex and exhibits paramagnetic behaviour?

(1) $[Cr(NH_3)_6]^{3+}$

(2) $[Co(NH_3)_6]^{3+}$

(3) $[FeF_6]^{3-}$

(4) $[Co(H_2O)_6]^{3+}$

**72.** Select the incorrect statement

(1) $KAl(SO_4)_2 \cdot 12H_2O$ is not a complex salt.

(2) The central atoms in complex compounds are also referred as Lewis acids.

(3) $[Co(NH_3)_4Cl_2]^{+}$ is a homoleptic complex.

(4) Ethylenediamine is didentate ligand.

**73.** Which of the following organometallic compounds will have weakest C – O bond?

(1) $[Co(CO)_4]^{-}$

(2) $[Ni(CO)_4]$

(3) $[Mn(CO)_6]^{+}$

(4) $[Cr(CO)_6]$

**74.** The complex ion having minimum magnitude of $\Delta_o$ (CFSE) in octahedral field is

(1) $[CoCl_6]^{3-}$

(2) $[Co(CN)_6]^{3-}$

(3) $[Co(H_2O)_6]^{3+}$

(4) $[Co(NH_3)_6]^{3+}$

**75.** Value of CFSE for high spin $d^5$ octahedral configuration will be

(1) $-2.0\Delta_0 + 2P$

(2) $-1.6\Delta_0 + P$

(3) Zero

(4) $-2.4\Delta_0 + 3P$

**76.** Which actinoid shows only one oxidation state?

(1) Th

(2) U

(3) Np

(4) Am

**77.** A : Sodium dichromate is more soluble than potassium dichromate in water.
R : Potassium dichromate is prepared by treating the solution of sodium dichromate with potassium chloride.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

**78.** The electronic configuration of Gadolinium (Atomic number = 64) is

(1) $[Xe]\ 4f^6, 5d^3, 6s^1$

(2) $[Xe]\ 4f^8, 5d^0, 6s^2$

(3) $[Xe]\ 4f^6, 5d^2, 6s^2$

(4) $[Xe]\ 4f^7, 5d^1, 6s^2$

**79.** Consider the following statements about interstitial compounds
(i) They are chemically inert
(ii) They are very hard
(iii) They retain metallic conductivity
The correct statements are

(1) (i) and (ii) only

(2) (ii) and (iii) only

(3) (i) and (iii) only

(4) (i), (ii) and (iii)

**80.** Which of the following is not a transition element?

(1) Sc

(2) Zn

(3) Cu

(4) Mn

**81.** Lanthanoids ion of largest size among the following is (Atomic numbers: Ce = 58, Nd = 60, Sm = 62, Yb = 70)

(1) $Ce^{3+}$

(2) $Nd^{3+}$

(3) $Sm^{3+}$

(4) $Yb^{3+}$

**82.** Among the following, second ionisation energy is maximum for

(1) V

(2) Ti

(3) Cr

(4) Mn

**83.** Which of the following is red in colour?

(1) Acidic $K_2Cr_2O_7$

(2) $K_2Cr_2O_7$ in basic medium

(3) $KMnO_4$

(4) $Cu_2O$

**84.** Which of the following exhibits only one type of lattice structure?

(1) Cr

(2) Fe

(3) Ti

(4) V

**85.** $HgCl_2$ reacts with excess of KI to give a red precipitate A which becomes soluble in excess of KI to form compound B. A and B are respectively

(1) $HgI_2$ & $K_2[HgI_4]$

(2) $HgI_2$ & $Hg_2I_2$

(3) HgI & $K[HgI_3]$

(4) HgI & $K_2[HgI_4]$

**86.** Compound which is not yellow in colour in the following is

(1) CdS

(2) $SnS_2$

(3) $BaCrO_4$

(4) HgS

**87.** Correct order of densities is

(1) Hg > Zn > Cd

(2) Hg > Cd > Zn

(3) Cd > Hg > Zn

(4) Zn > Cd > Hg

**88.** The metal present in Wilkinson catalyst is

(1) Rh

(2) Co

(3) Ir

(4) Tc

**89.** Ground state electronic configuration of Ce is (Atomic number of Ce = 58)

(1) [Xe] $4f^2\ 5d^0\ 6s^1$

(2) [Xe] $4f^1\ 5d^1\ 6s^2$

(3) [Xe] $5d^1\ 6s^2$

(4) [Xe] $4f^2\ 5d^1\ 6s^1$

**90.** In acidic medium, the colour of $CrO_4^{2-}$ changes into

(1) Orange

(2) Yellow

(3) Purple

(4) Green

**91.** If (+) - mandelic acid has a specific rotation of +150°, then the observed specific rotation of mixture containing 25% (–) - mandelic acid and 75% (+) - mandelic acid will be

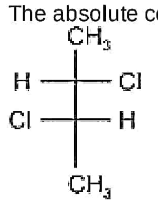
(1) +100°

(2) –100°

(3) +75°

(4) –75°

**92.** The absolute configuration of the following is

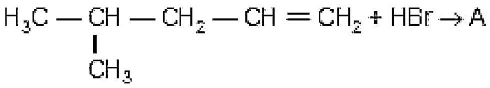



(1) 2S, 3R

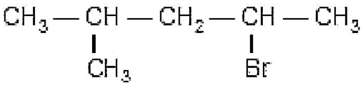
(2) 2S, 3S

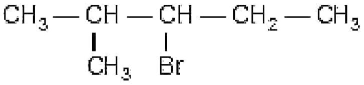
(3) 2R, 3S

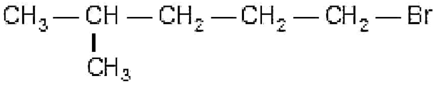
(4) 2R, 3R

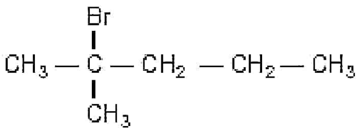
**93.** 

A (predominantly) is

(1) 

(2) 

(3) 

(4) 

**94.** 2-Bromo-3-methylpentane is heated with potassium ethoxide in ethanol. The major product obtained is

(1) 3-Methylpent-2-ene

(2) 2-Ethoxy-3-methylpentane

(3) 2-Methylpent-2-ene

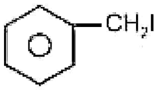
(4) 4-Methylpent-2-ene

**95.** Which compound is least likely to give yellow precipitate with $AgNO_3$(aq)?

(1) $CH_2 = CH - CH_2 - I$

(2) $CH_2 = CH - I$

(3) $CH_3 - CH_2 - CH_2 - I$

(4) 

**96.** Product C in the following reaction series is

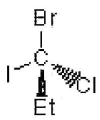
$H_3C - Br \xrightarrow{KCN} A \xrightarrow{H_3O^{\oplus}} B \xrightarrow{LiAlH_4} C$

(1) Acetaldehyde

(2) Ethanol

(3) Acetone

(4) Methane

**97.** The configuration of the stereocenter in  is

(1) R

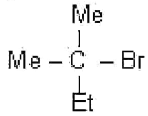
(2) S

(3) E

(4) Z

**98.** Among the following halides, the one that will undergo both $S_N1$ and $S_N2$ reaction rapidly?

a. $CH_3 - Br$

b. 

c. 

d. 

(1) a only

(2) c & d only

(3) b only

(4) a & d only

**99.** Total number of isomeric products (including stereoisomers) obtained when 2-chlorobutane is heated with alcoholic KOH is

(1) One

(2) Two

(3) Three

(4) Four



**100.** Which can act ambident nucleophile?

(1) $OH^-$

(2) $NH_3$

(3) $CN^-$

(4) $Cl^-$